Neuropsychologia 179 (2023) 108448

Contents lists available at ScienceDirect

Neuropsychologia

journal homepage: www.elsevier.com/locate/neuropsychologia

(Lack of) neural efficiency related to general giftedness and mathematical excellence: An EEG study

Ilana Waisman [a], Clemens Brunner [b], Roland H. Grabner [b], Mark Leikin [a], Roza Leikin [a,*]

[a] Faculty of Education, RANGE Center, University of Haifa, Israel
[b] Institute of Psychology, University of Graz, Austria

ARTICLE INFO

Keywords:
Neural efficiency
ERD/ERS
Learning-based task
Representations of functions
Excellence in mathematics
Giftedness

ABSTRACT

Previous studies on intelligence have demonstrated that higher abilities are associated with lower brain activation, indicating a higher neural efficiency. In other words, more able individuals use fewer brain resources. However, it is unclear whether the neural efficiency phenomenon also appears for mathematical performance, which is influenced by both domain-general giftedness and domain-specific competencies. Therefore, this study examined the effects of general giftedness (G) and excellence in mathematics (EM) on performance and brain activation while solving learning-based mathematical tasks that required translation from graphical to symbolic representations of functions. Overall, 118 high school students (aged 16–18) participated in the present study and were divided according to G and EM using a 2 × 2 study design. Participants worked on a function task requiring translation between symbolic and graphical representations of functions. Analyses of the behavioral data revealed positive effects of both G and EM on the accuracy of solutions and an interaction effect of both factors on reaction times, reflecting a positive effect of EM only among the gifted individuals. EEG analyses focused on oscillatory activity in the theta and alpha frequency bands and showed a significant effect of EM in the upper alpha band (10–12 Hz) event-related desynchronization (ERD) for both graphical and symbolic representations. Specifically, higher (compared to lower) EM was associated with a larger alpha ERD, indicating a higher level of brain activity. This stands in contrast with the neural efficiency phenomenon. These findings suggest that the neural efficiency phenomenon cannot be generalized to higher-order mathematical demands in high-performing individuals. Several explanations for this limitation are offered.

1. Introduction

In the research on the neurocognitive foundations of mathematical abilities and competencies, the focus has been on the lower to average performance range and on comparably simple mathematical tasks. Specifically, most of the previous neuroimaging studies have administered basic numerical tasks (e.g., drawing on magnitude processing) in individuals who show either an atypical development in terms of learning difficulties or a typical development with performance in the "normal" range. Much less is known, in contrast, about the neurocognitive mechanisms underlying excellent mathematical performance and the processes involved in more demanding tasks that are typical of high school mathematics instruction. This could partly be due to the challenge linked to the fact that high mathematical performance may be influenced by both domain-general cognitive ability (general giftedness or intelligence) and domain-specific mathematical competencies (mathematical excellence), which can be expected to have distinct neurofunctional correlates. In the present electroencephalography (EEG) study, we compared high school students with medium and high general cognitive abilities as well as medium and high mathematical competencies to provide further insight into the neurophysiological correlates of both factors. We administered a task drawing on mathematical functions that is representative of the mathematical demands in high school.

The terms 'mathematical giftedness', 'high mathematical abilities', 'high mathematical potential', and 'excellence in mathematics' are often considered synonymous terms (Leikin, 2020). In turn, 'mathematical talent' is considered a realized giftedness (Leikin, 2020). While mathematical giftedness is easy to distinguish in adults (for example, those who are research mathematicians and especially Fields Medal laureates), defining mathematical giftedness in students, and identifying such students, is not so clear. Some studies have referred to high

* Corresponding author.
E-mail addresses: waisman.ilana@gmail.com (I. Waisman), rozal@edu.haifa.ac.il (R. Leikin).

https://doi.org/10.1016/j.neuropsychologia.2022.108448
Received 7 April 2022; Received in revised form 10 December 2022; Accepted 13 December 2022
Available online 15 December 2022
0028-3932/© 2022 Elsevier Ltd. All rights reserved.

achievements in school mathematics as indicating mathematical giftnedness (Threlfall and Hargreaves, 2008; Koshy et al., 2009), some consider SAT-M scores (Prescott, Gavrilescu, Cunnington, O'Boyle and Egan, 2010; Lubinski and Benbow, 2006), whereas others focus on university mathematical majors (Amalric and Dehaene, 2016; Chi et al., 1981) or students with high IQs (Davidson and Sternberg, 1984; Popescu et al., 2019; Taub et al., 2008).

Leikin and Leikin (2009) suggested distinguishing between general giftedness and mathematical excellence and hypothesized that mathematical giftedness is a combination of both. To examine this hypothesis, they introduced a 2 × 2 design consisting of the factors general giftedness (G) and excellence in school mathematics (EM), and conducted a large-scale multidimensional study in 17–18 year old students with different combinations of EM and G levels (Leikin and Lev, 2013; Leikin et al., 2013a; Leikin et al., 2013b; Waisman et al., 2014; Paz-Baruch et al., 2014; Paz-Baruch et al., 2016; Leikin et al., 2016). The studies revealed that even though there is a correlation between EM and G factors, they are different in nature. For example, G was related to high verbal short-term memory, while EM was associated with high visuo-spatial working memory (Leikin et al., 2013). G-EM (gifted and excelling in mathematics) individuals excelled in visual-serial processing, while pattern-recall processing was found to be related to the G factor only (Paz-Baruch et al., 2016).

In contrast to mathematical excellence, there has been a vast amount of research on the neurophysiological correlates of general cognitive abilities or intelligence. This research has revealed that individual differences in intelligence are associated with activation differences in a wide-spread network of brain regions (Colom et al., 2010; Deary et al., 2010; Jung and Haier, 2007). A well-established finding in this research is the neural efficiency phenomenon, indicating that "brighter individuals display lower (more efficient) brain activation while performing cognitive tasks" (Neubauer and Fink, 2009, p. 1004). This result pattern was observed for a variety of different cognitive tasks, from rather simple reaction time tasks to more demanding working memory tasks (for a review, cf. Neubauer and Fink, 2009).

Most of the studies demonstrating the neural efficiency phenomenon analyzed induced (oscillatory) EEG activity in the (higher) alpha frequency band (about 10–12 Hz) while solving cognitive tasks (for a review, cf. Neubauer and Fink, 2009). Induced EEG activity indicates task-related coupling and uncoupling of functional networks in the brain (Bastiaansen and Hagoort, 2003; Klimesch et al., 2005) and can be measured with event-related (de-) synchronization (for a review, cf. Pfurtscheller and Da Silva, 1999). In the alpha frequency band, the amount of alpha power typically decreases during cognitive activity as compared to a resting state, a phenomenon which is labelled event-related desynchronization (ERD). The percentage amount of decrease is then expressed in the ERD measure. In the studies on the neural efficiency phenomenon, higher intelligence was associated with lower alpha ERD, which has been interpreted to reflect lower brain activation during task performance (Grabner et al., 2006; Grabner et al., 2004).

Given that higher general cognitive abilities are associated with more efficient brain activation, the question arises whether this can be generalized to domain-specific competencies or excellence. In a seminal study on chess players, Grabner et al. (2006) found that chess experts (compared to novices) displayed more efficient brain activation (lower alpha ERD) while solving two domain-related tasks drawing on mental speed (enumeration of specific chess pieces) and reasoning (finding a checkmate move). This expertise-related difference was also independent of their intelligence level. However, in a task requiring short-term memory of chess positions, during both encoding and retrieval experts displayed less efficient brain activation (higher alpha ERD) compared to novices.

In the domain of mathematics, most studies on induced EEG activity have focused on arithmetic problem solving and examined both the alpha and the theta (4–7 Hz) frequency band. Specifically, it has been observed that solving easy vs. complex arithmetic problems is associated with higher event-related synchronization (ERS; task-related bandpower increase) in the theta band and lower ERD in the alpha band (De Smedt et al., 2009; Grabner and De Smedt, 2011; Grabner et al., 2022). This result pattern has been explained as reflecting the application of different cognitive strategies to solve arithmetic problems of different complexity levels. Easy problems (such as single-digit multiplications) have been found to be typically solved through arithmetic fact retrieval (i.e., retrieval of the solution from long-term memory). Since theta ERS has been assumed to link lexical-semantic representations over the cortex and between the cortex and the hippocampus (e.g., Bastiaansen and Hagoort, 2006; Marko et al., 2019; Salisbury and Taylor, 2012), higher theta ERS while solving easy (compared to more complex) arithmetic problems was interpreted to reflect a larger reliance on arithmetic fact retrieval. The higher alpha ERD in complex as compared to easy problems was interpreted to reflect the application of effortful and attention-demanding procedures, as alpha ERD has been found to increase with task difficulty and the associated increase in demand on attentional processes (Klimesch, 2012; Ku et al., 2010; Pfurtscheller and Da Silva, 1999).

To our knowledge, induced EEG correlates of individual differences in mathematical competencies have only been investigated in one study so far. Artemenko et al. (2019) divided a sample of adults into two groups with different mathematical abilities based on their performance on an arithmetic task. During EEG recording, participants were required to solve multiplication and division problems of two complexity levels. Similar to previous studies on induced EEG activity during arithmetic problem-solving, the solution of more complex problems was associated with lower theta ERS and higher alpha ERD. This complexity effect, however, was not modulated by the participants' mathematical ability, indicating that the two groups showed similar complexity effects during problem-solving.

Even though arithmetic problems are an essential foundation for mathematical skills, they are not representative of mathematical demands in high school. A key concept in the high school curriculum is mathematical functions (Da Ponte, 1992; Leinhardt et al., 1990). Understanding of functions includes comprehension of functional relation (for example as a particular type of correspondence between two sets of objects) and knowledge and skills related to particular types of functions such as linear, quadratic, polynomial, rational, trigonometric, exponential, or logarithmic functions. The concept of a function is the basis of the concepts of derivative and integral as well as of the theory of limits.

Mathematical functions require the competence to translate and connect between different representations of mathematical concepts (Hiebert and Carpenter, 1992). Kaput (1998) argued that the sources of building mathematical meaning are found in the translations between representation systems. According to Duval (2006), translations between representations are "at the heart of mathematical activity" (Duval, 2006, p. 107). The learners' competence to translate flexibly between different representations allows them to comprehend the multifaceted nature of various concepts (Ainsworth et al., 1997; Cox, 1999; Elia et al., 2007; Gagatsis and Shiakalli, 2004; Schönborn and Bögeholz, 2009; Van Heuvelen and Zou, 2001). This competence is built up in mathematics instruction; therefore, mathematical function tasks can be considered learning-based tasks. Specifically, learning-based tasks involve implementation of procedures, algorithms, strategies and representations that were studied and trained in previous learning experiences. These tasks are often contrasted with insight-based tasks, which are unfamiliar to students and require problem restructuring to allow the use of knowledge and skills in a new situation.

An EEG study on the mathematical function task was conducted by Waisman et al. (2014). They examined the interplay of G and EM factors on high school students' mathematical performance (75 students, 16–18 years old) associated with translations from graphical to symbolic representations of functions using evoked EEG activity (i.e., event-related potentials; ERPs). Main effects of EM factor appeared at the

behavioral and electrophysiological levels, while main effects of G factor were found at the behavioral level only. When solving the function tasks, the accuracy of responses in EM (excelling in mathematics) students was significantly higher than the accuracy of NEM (non-excelling in mathematics) students, and the accuracy in G (gifted) students was significantly higher than in their NG (non-gifted) counterparts. At the neurophysiological level, EM students exhibited significantly higher ERP amplitudes in the 300–500 ms interval as compared to NEM students.

The reported ERP findings have provided important first insights into the electrophysiological correlates of G and EM while solving learning-based tasks representative of high school mathematics. However, their significance with respect to the neural efficiency hypothesis is limited, as the relationship between ERP amplitudes and intelligence is less well established than the association between alpha oscillations and intelligence. While it is an often replicated finding that more intelligent individuals display lower alpha ERD during cognitive tasks, inconsistent results have been observed for ERP amplitudes (cf. Neubauer and Fink, 2009). In addition, ERPs represent short phase-locked responses to a certain stimulus that do not capture the problem-solving process in more demanding cognitive tasks (e.g., after half a second), whereas ERD/ERS can flexibly measure induced activity from the problem's onset to its solution. Therefore, the aim of the present study was to evaluate the roles of G and EM on neural efficiency using the well-established and frequently used method of ERD/ERS. Since both theta and alpha frequency bands have been associated with mathematical processes in previous studies, we conduct analyses for both bands.

1.1. Research hypotheses

In line with the above-noted previous findings (Waisman et al., 2014), we examine the following hypotheses.

H1. Both G and EM factors have a significant effect on behavioral measures when solving functions tasks. A. EM students exhibit higher accuracy and shorter reaction times than NEM students. B. G students show higher accuracy and shorter reaction times than NG students.

H2. Due to the learning-based nature of function tasks, a pattern of neural efficiency can be observed only for the EM factor: EM students show lower brain activation (as reflected in lower alpha ERD) as compared to NEM students in S1 (processing of function) and S2 (verification of equation).

No specific hypotheses regarding theta ERD/ERS can be proposed as there are no previous studies relating this measure to intelligence and mathematical excellence. Based on Artemenko et al. (2019), no group differences can be expected.

2. Methods

2.1. Research sample

In this study we report on a group of 118 students comprised of 45 females and 73 males with a mean age of 16.6 (standard deviation 0.6) years. These participants were chosen based on a sampling procedure aimed at forming groups that differed in general giftedness (G factor) and excellence in mathematics (EM factor) (Table 1). For this purpose, we examined 1200 Israeli students in 10th and 11th grades (16–18 years old) using the Raven's Advanced Progressive Matrices Test (RPMT) (Raven et al., 2000) and the Scholastic Assessment Test in Mathematics (SAT-M) (Benbow and Lubinski, 1996).

The *EM factor* was determined by the level of mathematics the students studied in school and their scores on the SAT-M test. Mathematics is a compulsory subject in Israeli high schools and students are placed in one of three levels of mathematics: high (HL), regular (RL), and low (LL) level. The level of instruction is determined by students' mathematical achievements in earlier grades. Instruction at HL differs from that at RL in terms of the depth of the learning material and the complexity of the mathematical problem-solving involved. Overall, roughly 15% of students in 10th, 11th and 12th grades study mathematics at HL. Students with mathematics scores above 90 (maximum of 100; scored by mathematics teacher) in HL mathematics represent approximately 3% of the general population. SAT-M is a multi-component test that includes tasks from different domains of mathematics. It is a recognized tool for measuring academic abilities in mathematics (e.g., Young and Kobrin, 2001) and strategic problem-solving skills (e.g., Morrison et al., 2006). We used a short version of the SAT-M that contained 35 items with a time limit of 30 min (adopted by Zohar, 1990). The range of correct scores could vary from 0 to 35 and represented the number of problems solved correctly. A SAT-M score above 26 was chosen as a control measure for the EM sampling (2% of the general population). Thus, students were assigned to the EM group when meeting two criteria: a score in HL mathematics instruction above 90 and an SAT-M score above 26. The NEM group included students with an SAT-M score less than 22 (of 35) and a mathematics score greater than 90 in RL math or lower than 80 in HL math. An SAT-M score of 22 and HL math score below 80 were used as a cut-off to ensure that the groups clearly differed in their mathematical achievements.

Table 1
Distribution of the participants among the study groups.

	Gifted (G) RPMT >27<	Non-Gifted (NG) RPMT ≤26	Total
Excelling in mathematics (EM) SAT-M > 26 of 35 and score >90 in HL math	25	33	58
Non-Excelling in mathematics (NEM) SAT-M < 22 of 35 and score >90 in RL math **or** score <80 in HL math	23	37	60
Total	48	70	118

SAT-M - Scholastic Assessment Test in Mathematics; RPMT Raven's Advanced Progressive Matrices TestG - gifted, NG - non-gifted, EM - excelling in mathematics, NEM-non-excelling in mathematics.

The *G factor* was determined by students' participation in gifted classes and their success on the RPMT test. Students assigned to the G group were mainly chosen from classes for gifted students (identified by a national examination as having IQ > 130 in the third grade). The students in classes for gifted students represent 2% of the population in Israel. Additionally, we used the RPMT test, which is considered a good indicator of general intelligence and ability (Carpenter et al., 1990; Silverman, 2009). We used a shortened test version containing 30 items (adopted by Zohar, 1990) so that the range of the final score (representing the number of problems solved correctly) could vary from 0 to 30. The test time limit was 15 min. We used this test for two purposes: (1) for the validation of the G factor for students from gifted classes, which initially is determined by the ministry of education in the 3rd grade using IQ tests, and (2) for inclusion of students from regular classes in the G group, since when the classes are formed in the second grade students are allowed but not obliged to study in gifted classes. Thus, there is a possibility that G students can be found in classes that are not denoted as classes for gifted students. Raven scores of higher than 27 served as a criterion for students' inclusion in the G group. After the sampling procedure, students were divided into four experimental groups designed with varying combinations of the EM and G factors (see Table 1):

- G–EM group ($N = 25$; 20 males, 5 females): Students who are identified as generally gifted and excelling in mathematics.
- G–NEM group ($N = 23$; 22 males, 1 female): Students who are identified as generally gifted but do not excel in mathematics.

- NG–EM group ($N = 33$; 14 males, 19 females): Students who are not identified as generally gifted but excel in mathematics.
- NG–NEM group ($N = 37$; 16 males, 21 females): Students who are neither identified as generally gifted nor do they excel in mathematics.

All participants were paid volunteers and were native speakers of Hebrew, right-handed, with no history of learning disabilities and/or neurological disorders, and who had normal or corrected to normal vision. All participants and their parents signed an informed consent form. The study received the approval of the Helsinki Committee (Rambam Medical Center), the Israel Ministry of Education, and the Ethics Committee of the University of Haifa.

2.2. Materials and procedure

A computerized test with 60 trials was designed using E-Prime software (Schneider et al., 2002). The test required participants to translate between graphical (depicted in stage 1; S1) and symbolic (depicted in stage 2; S2) representations of functions (Fig. 1). 30 of the 60 trials contained a correct answer, while the other 30 trials contained an incorrect answer. The task design was based on the 4-strategies theory of problem-solving process identified by Polya (1973): understanding the problem, devising a plan, carrying out the plan, and looking back. The design of each trial included two stages: S1 – problem condition – corresponding to understanding the problem and devising a plan, and S2 – suggested answer – corresponding to Polya's looking back strategy. Carrying out a plan could be performed either at S1 or S2 (Waisman et al., 2014)

All trials were presented visually at the center of the computer screen, in black characters on a white background (Fig. 1). Participants received a graph of a mathematical function shown for 2000 ms followed by an equation. They had to determine whether the graph and the equation represented the same function by pressing the button "3" (correct) or "1" (incorrect) according to their decision. The equation remained visible until the participant responded or for a maximum of 5000 ms. The test was proved to be reliable with Cronbach's alpha of .859.

2.2.1. EEG recording

We continuously recorded the electroencephalogram (EEG) using a 64-channel BioSemi ActiveTwo system (BioSemi, Amsterdam, The Netherlands) and ActiveView recording software. Pin-type electrodes were mounted on a customized Biosemi head-cap arranged according to the extended 10–20 system. Two flat electrodes were placed on the sides of the eyes to monitor horizontal eye movement. A third flat electrode was placed underneath the left eye to monitor vertical eye movement and blinks. During the session, electrode offsets were kept below 50 μV. EEG signals were amplified and digitized with a 24-bit AD converter with a sampling frequency of 2048 Hz and a low-pass filter at 100 Hz.

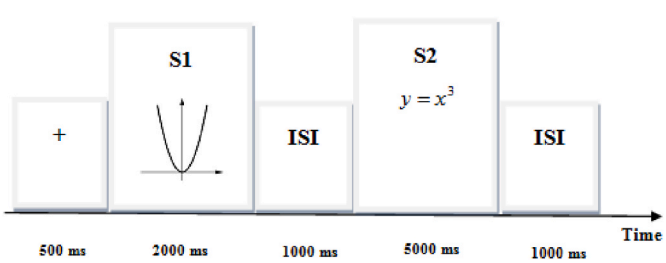

Fig. 1. The sequence of events in a trial.

2.3. Data analysis

2.3.1. Behavioral data

Analyses were applied to accuracy (Acc) and reaction time for correct responses (RTc). Acc was determined by the percentage of participants' correct responses to all 60 trials on the test. RTc was calculated as the mean time spent for verification of an answer (stage S2) in all correctly solved trials. We analyzed differences in Acc and RTc using analysis of variance (ANOVA) for G (gifted vs. non-gifted) and EM (excelling vs. non-excelling in mathematics) as between-subject factors.

2.3.2. EEG data

EEG data were preprocessed offline using the Brain Vision Analyzer software (Brain Products). First, EEG data were bandpass-filtered between 0.53 and 30 Hz (zero-phase) and referenced to the common average of all electrodes. We divided the continuous EEG data into short non-overlapping epochs of 400 ms duration for the purpose of removing artifacts. We discarded all epochs containing signals exceeding ±80 μV on any channel. Ocular artifacts were corrected using a regression-based approach (Gratton et al., 1983). Finally, EEG data were downsampled to 256 Hz. Using the preprocessed data, we computed ERD/ERS separately for two frequency bands (theta: 4–7 Hz; higher alpha: 10–12 Hz) during the S1 phase (presentation of the graph) and during the S2 phase (presentation of the equation). We used the median of the 500 ms fixation period as baseline R, and the median of the presentation time window as activity A to compute $ERD/ERS = \left(\frac{A}{R} - 1\right)\cdot 100\%$

In this calculation, we used only trials with correct responses and reaction times of at least 600 ms, grouped by trial type (there were 30 trials with a correct answer and 30 trials with an incorrect answer). We selected the 600 ms cutoff to exclude trials with unusually short response times, since these likely do not represent valid responses. Furthermore, since trial duration is inversely proportional to the lowest frequency that can be analyzed, we selected this threshold to ensure that we could still compute theta band ERD/ERS. Afterwards, we averaged the channel data into the following nine regions of interest:

1. Anterior left: AF7, AF3, F7, F5, F3
2. Anterior middle: F1, FZ, AFZ, F2
3. Anterior right: AF4, AF8, F4, F6, F8
4. Central left: FT7, FC5, FC3, C3, C5, TP7, CP5, CP3
5. Central middle: FC1, C1, CP1, FCZ, CZ, CPZ, FC2, C2, CP2
6. Central right: FC4, FC6, FT8, C4, C6, CP6, CP4, TP8
7. Posterior left: P3, P5, P7, PO7, PO3, O1
8. Posterior middle: P1, PZ, P2, OZ, POZ
9. Posterior right: P4, P6, P8, PO8, PO4, O2

The regions 1 to 9 are grouped into two factors, caudality (anterior, central, posterior) and laterality (left, middle, right), in subsequent analyses.

Repeated measures ANOVAs were computed for the theta and alpha frequency band separately, taking G (gifted vs. non-gifted) and EM (excelling vs. non-excelling) as between-subjects factors and Caudality (anterior, central and posterior) and Laterality (left and right) as within-subjects factors.

For all analyses, p-values were corrected for deviation from sphericity according to the Greenhouse-Geisser method. Following a significant interaction, post-hoc tests using the Bonferroni adjustment were performed.

3. Results

3.1. Acc and RTc in four groups of participants

The mean Acc and RTc are displayed in Table 2. The G factor and EM factor had significant effects on Acc (Fig. 2a). Acc in G students was

Table 2
Descriptive statistics of Acc and RTc in the four groups of participants.

Measure		Mean (SD)		
		G	NG	**Overall**
Acc (%)	EM	82.6 (7.4)	76.6 (11.5)	79.2 (10.3)
	NEM	79.9 (8.6)	71.4 (12.2)	74.7 (11.6)
	Overall	81.3 (8.0)	73.9 (12.1)	
RTc (ms)	EM	1566.2 (376.1)	1759.0 (404.1)	1676.0 (425.8)
	NEM	1854.7 (418.4)	1690.2 (423.6)	1753.2 (425.8)
	Overall	1704.5 (418.8)	1722.6 (413.0)	

$^{*}p \leq .05$, $^{**}p \leq .01$, $^{***}p \leq .001$ Acc – Accuracy; RTc – Reaction time for correct responses.

significantly higher [$F(1,114) = 13.617$, $p < .001$, $\eta_p^2 = 0.107$] compared to that in NG students.

In the RTc (Fig. 2b), we only found a significant interaction of G and EM factors [$F(1,114) = 5.460$, $p = .021$, $\eta_p^2 = 0.046$]. The post-hoc pairwise comparisons demonstrated that G-EM students had significantly lower RTc [$F(1,114) = 6.002$, $p = .016$, $\eta_p^2 = 0.050$] than G-NEM students while NG-EM students had similar RTc as their NG-NEM counterparts.

3.2. Electrophysiological findings in four groups of participants

3.2.1. Findings at S1

Fig. 3 demonstrates significant differences in the alpha band linked to EM and G factors.

In the alpha band, we found a significant main effect of EM factor [$F(1,114) = 4.742$, $p = .031$, $\eta_p^2 = 0.040$] and a significant interaction of G factor with EM factor [$F(1,114) = 5.995$, $p = .016$, $\eta_p^2 = 0.050$, see Fig. 3a]. EM students elicited significantly higher alpha ERD (i.e., more negative values) than their NEM counterparts (−23.28 vs. −15.13, respectively). Further pairwise comparison demonstrated that G-EM students showed significantly higher alpha ERD than NG-EM students [$F(1,114) = 5.289$, $p = .013$, $\eta_p^2 = 0.053$] and than G-NEM students [$F(1,114) = 9.024$, $p = .003$, $\eta_p^2 = 0.073$], while NG-EM and NG-NEM elicited similar ERD.

Additionally, we found a significant interaction of Caudality with G and EM factors [$F(2, 228) = 5.135$, $p = .007$, $\eta_p^2 = 0.043$]. Post hoc tests in each caudality site revealed significant interactions of G and EM factors in the central [$F(1,114) = 7.537$, $p = .007$, $\eta_p^2 = 0.062$] and in the posterior [$F(1,114) = 7.966$, $p = .006$, $\eta_p^2 = 0.065$] electrode sites (Fig. 3b). The result pattern in the central and posterior sites is similar to the overall interaction between G and EM factors depicted in Fig. 3a.

Finally, we found a significant interaction of Laterality with G and EM factors [$F(2,228) = 3.838$, $p = .023$, $\eta_p^2 = 0.033$]. Post hoc tests in each laterality site found that the significant interaction of G and EM factors appeared in the left [$F(1,114) = 8.525$, $p = .004$, $\eta_p^2 = 0.070$] and in the middle [$F(1,114) = 6.634$, $p = .011$, $\eta_p^2 = 0.055$] electrodes sites (Fig. 3c). Again, in the areas with significant interactions, the result pattern is similar to the overall interaction between G and EM.

In the theta frequency band, no significant effects emerged.

3.2.2. Findings at S2

In the alpha band, we found a main effect of EM factor [$F(1,114) = 4.145$, $p = .044$, $\eta_p^2 = 0.035$]. Like at S1, EM students elicited significantly higher alpha ERD than their NEM counterparts (−32.0 vs. −24.3, respectively). No other main effects or interactions emerged in the alpha band.

In the theta band, we did not find any significant main effect or interaction.

4. Discussion

This study aimed at examining the effects of general giftedness and mathematical excellence on oscillatory EEG patterns associated with a learning-based task requiring translation between different representations of a function. Following the 2 × 2 design proposed by Leikin and Leikin (2009) we examined the effects of G and EM on accuracy and reaction times and used ERD/ERS methodology for alpha and theta frequency bands.

We expected that EM students would exhibit higher accuracy and shorter reaction times than NEM students since functions are a central concept in their school mathematical curriculum (Hypothesis 1A). Our hypothesis was confirmed with respect to accuracy, but no significant main effect of the EM factor on reaction times was found. This result pattern replicates findings reported in Waisman et al. (2014), who also observed a main effect of EM only on accuracy but not on reaction times. There are only slight differences in mean values, which may be traced back to including female participants in the current study as contrasted to a purely male sample in the previous publication. These findings indicate that students excelling in mathematics outperform their non-excelling peers in the number of correct solutions but not in the amount of time needed to verify whether an equation matches a graphical function. Ainsworth (1999) contends that fluent translation across different representations is an essential characteristic of

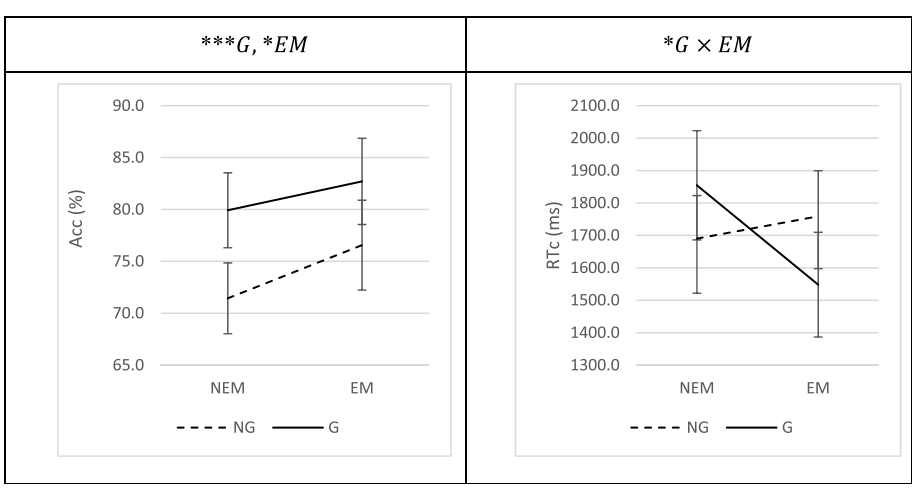

Fig. 2. (a) Acc (left panel) and (b) RTc (right panel) in the four groups of participants. The error bars represent the 95% CI error of the mean. Above the graphs, the significant effects are depicted. $^{***}p < .001$; $^{*}p < .05$.

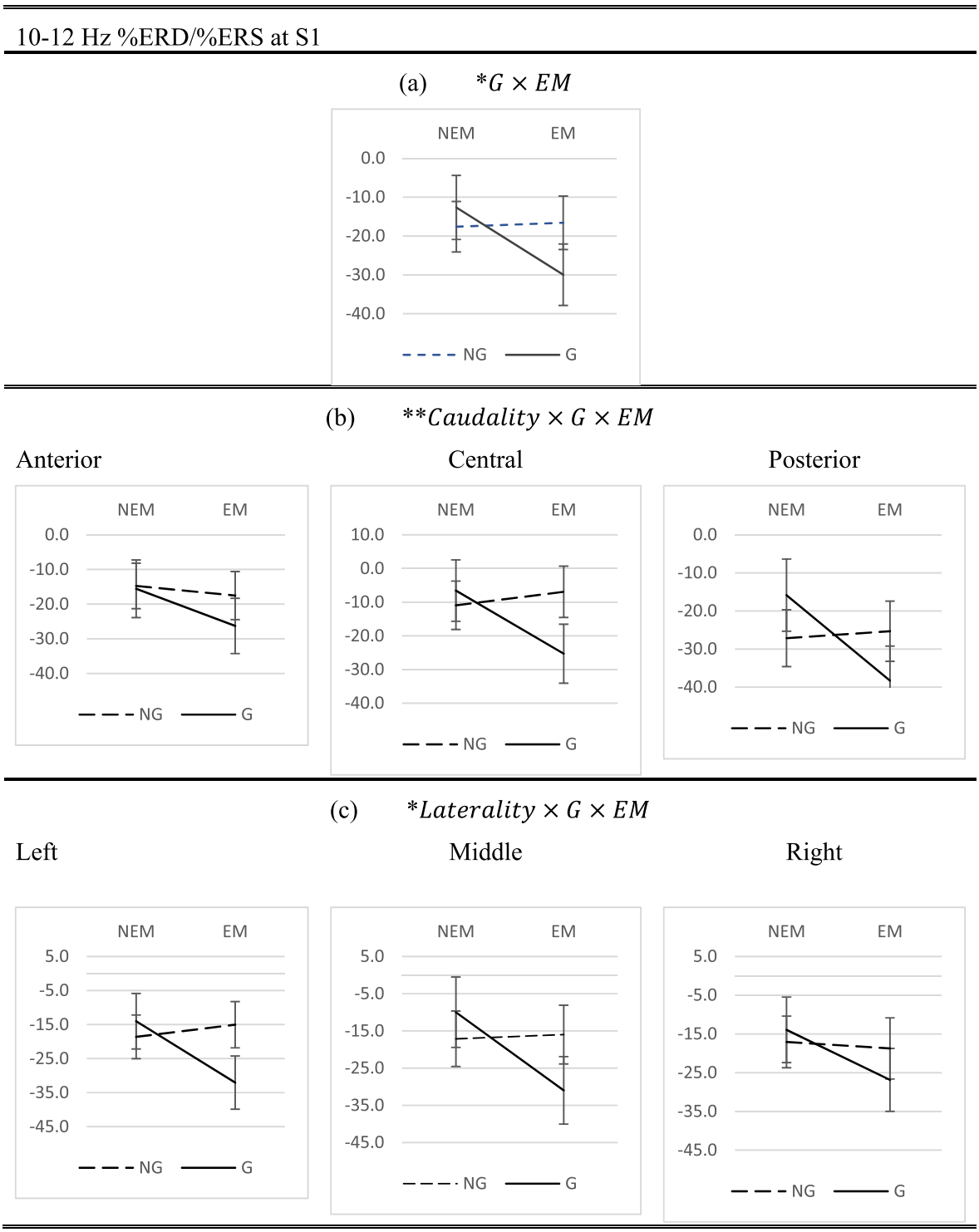

Fig. 3. Significant differences in the event-related (de-)synchronization (%ERD/ERS) in the alpha band (10–12 Hz) at S1 linked to EM and G factors. The error bars represent the 95% CI error of the mean. Above the graphs, the corresponding significant effects are depicted. $^{**}p < .01$; $^{*}p < .05$.

expertise.

Consistently with the findings of Waisman et al. (2014), we hypothesized that G students would also show higher accuracy than NG students (Hypothesis 1B), which was indeed observed in the present data. Similar to the EM factor, this general performance advantage was restricted to accuracy.

However, even though no main effects of EM and G emerged for the reaction times, there was a significant interaction of both factors. This interaction revealed that within the group of NG students the reaction times were independent of the level of EM. Only in the G students did we find faster responses in the EM compared to the NEM students, which is in line with Waisman et al. (2014). This finding extends those by Waisman et al. (2014), who only analyzed main effects, and suggests that when students possess a high level of general intelligence, their learning-related mathematical excellence has a beneficial impact on their speed of response.

On the neurophysiological level, we hypothesized a pattern of neural efficiency (i.e., lower alpha ERD in more able individuals) for the EM

factor (Hypothesis 2). This hypothesis was not supported. At both S1 and S2, the EM factor had a significant main effect on alpha ERD in that EM students elicited higher ERD than NEM students. Even though ERPs cannot be directly compared with alpha ERD, it can be argued that this unexpected result resembles that of Waisman et al. (2014) who found that at S1 EM students had larger late ERP amplitudes as compared to NEM students.

At S1, a significant interaction of G factor with EM factor was found. G-EM students showed significantly higher alpha ERD as compared to G-NEM students while NG-EM and NG-NEM elicited similar ERD. The interactions of Laterality with G and EM factors and Caudality with G and EM factors revealed that this result pattern was quite representative for different regions. Specifically, the interaction between G and EM factors was similar in the central and posterior sites and in the left and middle sites.

The alpha ERD results in S1 (given in Fig. 3a) appear to be similar to the interaction on reaction times (shown in Fig. 2b) but in the opposite direction. While lower reaction times reflect higher performance, higher alpha ERD (i.e., more negative values) is indicative of higher and, thus, less efficient brain activation. Those students with the best performance overall – the G-EM students – displayed the highest level of brain activity, while those with the worst performance – the NG-NEM students – showed a comparably low level of activation. In addition, within the NG students EM did not have an effect, whereas in the G students EM was associated with higher upper alpha ERD, the opposite of what we expected based on the neural efficiency phenomenon.

One possible explanation for the absence of neural efficiency in EM students is the learning-based nature of the administered task, i.e., the fact that it relies strongly on previously acquired knowledge. Neuroimaging studies on domain-specific competencies or expertise have reported that higher competencies or expertise are often related to higher instead of lower levels of brain activation (e.g., Kelly and Garavan, 2005). These have been attributed either to a larger reliance on task-related brain regions or the application of more sophisticated strategies in problem-solving. The higher activation in the EM students of the present sample (compared to their NEM peers) may be indicative of these processes. Unfortunately, due to a lack of EEG studies on the neurophysiological correlates of processes involved in solving the function task, no conclusion can be drawn as to whether they only differed in the amount of activation in task-related regions or also in regions associated with problem-solving strategies.

A related explanation considers the distinction between tasks focusing on short- or working-memory, on the one hand, and long-term memory, on the other hand. As elaborated in Neubauer and Fink (2009), positive relationships between alpha ERD and intelligence (i.e., contradicting the neural efficiency hypothesis) have been observed in tasks that require access to (semantic) long-term memory, indicating that "the involvement of LTM [long-term memory] demands could be a possible moderator variable in the brain-IQ relationship" (p. 1015). Although there is not yet clear evidence for this moderating variable, the findings of the present study add to this notion in that the successful solution of the function problems is strongly dependent on the students' prior knowledge stored in long-term memory.

Another explanation is linked to the moderating influence of task complexity and difficulty (Neubauer and Fink, 2009). That is, neural efficiency mainly arises when individuals are faced with tasks of low to moderate complexity/difficulty, such as semantic letter matching (Neubauer et al., 1999) or easy intelligence problems (Doppelmayr et al., 2005a,b). However, when the task complexity or difficulty is high, more intelligent individuals seem to devote more cortical resources as compared to less intelligent individuals, and thus exhibit higher brain activation (Dunst et al., 2014; Nussbaumer et al., 2015). In other words, the negative association between intelligence and brain activation level arises only for tasks that do not place particularly high demands on the participants. The function task administered in the present study can be regarded as a task with high demands. Learning functions is part of the algebra curriculum, and algebra has long been assumed to be a more complex domain of mathematics than, for instance, arithmetic (Booth, 1988; Eisenberg, 2002; Lee et al., 2017). In addition, translation between representations is thought to be difficult for learners (Adu-Gyamfi et al., 2017; Ainsworth, 1999; Janvier, 1987; Keig and Rubba, 1993). Thus, our findings for the higher ERDs in the upper alpha band in G-EM (compared to NG-NEM) students may reflect a larger amount of invested resources to solve the complex task accurately as fast quickly as possible. A limitation of this explanation, however, can be seen in the comparably high levels of accuracies (group averages between 71.4 and 82.6%), indicating that even though the task demands are complex, the task difficulty may not be very high. This raises the question of the distinctive effects of task complexity (in terms of processing demands) and task difficulty (in terms of solution rate) on neural efficiency, which appears to be a promising avenue for future research.

An additional possible explanation for the lack of neural efficiency associated with the EM factor at S1 is linked to the visual (graphical) representation in which functions were depicted for students. A recent study demonstrated that visual abilities do not predict mathematical giftedness, with mathematical giftedness defined as a combination of G and EM as in the G-EM group (Paz-Baruch et al., 2022). It could be speculated that G-EM students proceed immediately to symbolic representation at S1 without waiting until S2 due to their inclination to work with symbolic representations. This assumption is also in line with Clements (1984) who showed that mathematically gifted individuals are prone to deal with symbolic representations. On the other hand, since G-EM students showed significantly higher alpha ERD than the other three groups, we can assume that the latter groups did not translate the graphical representation to a symbolic representation at S1 but rather verified the correctness of the symbolic representation at S2.

In addition to the alpha band, we exploratively analyzed potential effects of G and EM on the theta frequency band, as this band turned out to be sensitive to mathematical problem-solving processes in previous EEG studies (De Smedt et al., 2009; Grabner and De Smedt, 2011; Grabner et al., 2022). Neither in S1 nor in S2 were any effects of G and EM observed in this frequency. Even though this is in line with the results reported by Artemenko et al. (2019), who administered an arithmetic task and failed to find a moderating effect of mathematical competence on theta (as well as alpha) ERD/ERS, effects of EM would have been plausible. Specifically, since higher theta ERS was found to be associated with a larger reliance on domain-specific knowledge (arithmetic facts) during problem solving, EM students (possessing a larger knowledge base to draw on) could have displayed higher theta ERS than NEM students. The lack of this effect suggests that the automatic fact retrieval processes in arithmetic that are related to theta ERS cannot be generalized to the retrieval of more complex (less factual and more conceptual) mathematical knowledge needed to solve mathematical function tasks.

Taken together, the present study is the first ERD/ERS study to investigate the neural correlates of the factors EM and G while solving a complex mathematical task representative of the high school curriculum. We found that both EM and G positively affected task accuracy and that both factors interacted with each other in the speed of problem-solving. The level of brain activation was strongly dependent on students' EM in that students with higher mathematical excellence displayed stronger brain activation in both stages of solving function problems. At the first stage of problem-solving, EM had an impact on brain activity primarily among the G students. The pattern of results was opposite to the neural efficiency hypothesis, possibly due to the characteristics of the function task (complex task drawing on domain-specific knowledge).

Credit author statement

Ilana Waisman: Investigation, Formal analysis, Writing; Clemens Brunner: Formal analysis, Writing – review & editing; Roland Grabner:

Conceptualization, Writing – review & editing,; Mark Leikin: Methodology, Project administration, Funding acquisition; Roza Leikin: Conceptualization, Methodology, Writing, Funding acquisition.

Data availability

Data will be made available on request.

Acknowledgement

This project was made possible through the support of a grant 1447 from the John Templeton Foundation. The opinions expressed in this publication are those of the author(s) and do not necessarily reflect the views of the John Templeton Foundation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.neuropsychologia.2022.108448.

References

Adu-Gyamfi, K., Bossé, M.J., Chandler, K., 2017. Student connections between algebraic and graphical polynomial representations in the context of a polynomial relation. Int. J. Sci. Math. Educ. 15 (5), 915–938.
Ainsworth, S., 1999. The functions of multiple representations. Comput. Educ. 33 (2–3), 131–152.
Ainsworth, S.E., Bibby, P.A., Wood, D.J., 1997. Information technology and multiple representations: new opportunities–new problems. J. Inf. Technol. Teach. Educ. 6 (1), 93–105.
Amalric, M., Dehaene, S., 2016. Origins of the brain networks for advanced mathematics in expert mathematicians. Proc. Natl. Acad. Sci. USA 113 (18), 4909–4917.
Artemenko, C., Soltanlou, M., Bieck, S.M., Ehlis, A.C., Dresler, T., Nuerk, H.C., 2019. Individual differences in math ability determine neurocognitive processing of arithmetic complexity: a combined fNIRS-EEG study. Front. Hum. Neurosci. 13, 227.
Bastiaansen, M., Hagoort, P., 2003. Event-induced theta responses as a window on the dynamics of memory. Cortex 39 (4–5), 967–992.
Bastiaansen, M.C.M., Hagoort, P., 2006. Oscillatory neuronal dynamics during language comprehension. In: Neuper, C., Klimesch, W. (Eds.), Progress in Brain Research, vol. 159. Elsevier, pp. 179–196.
Benbow, C.P.E., Lubinski, D.J.E., 1996. Intellectual Talent: Psychometric and Social Issues. Johns Hopkins University Press.
Booth, L.R., 1988. Children's difficulties in beginning algebra. In: Coxford, F., Schulte, A. P. (Eds.), The Ideas of Algebra, K-12: 1988 Yearbook. NCTM, Reston, pp. 43–60.
Carpenter, P.A., Just, M.A., Shell, P., 1990. What one intelligence test measures: a theoretical account of the processing in the Raven Progressive Matrices Test. Psychol. Rev. 97 (3), 404.
Chi, M.T., Feltovich, P.J., Glaser, R., 1981. Categorization and representation of physics problems by experts and novices. Cognit. Sci. 5 (2), 121–152.
Clements, M.A., 1984. Terence tao. Educ. Stud. Math. 15 (3), 213–238.
Colom, R., Karama, S., Jung, R.E., Haier, R.J., 2010. Human intelligence and brain networks. Dialogues Clin. Neurosci. 12 (4), 489–501.
Cox, R., 1999. Representation construction, externalised cognition and individual differences. Learn. InStruct. 9 (4), 343–363.
Davidson, J.E., Sternberg, R.J., 1984. The role of insight in intellectual giftedness. Gift. Child. Q. 28 (2), 58–64.
De Smedt, B., Grabner, R.H., Studer, B., 2009. Oscillatory EEG correlates of arithmetic strategy use in addition and subtraction. Exp. Brain Res. 195 (4), 635–642.
Deary, I.J., Penke, L., Johnson, W., 2010. The neuroscience of human intelligence differences. Nat. Rev. Neurosci. 11 (3), 201–211.
Doppelmayr, M., Klimesch, W., Sauseng, P., Hödlmoser, K., Stadler, W., Hanslmayr, S., 2005a. Intelligence related differences in EEG-bandpower. Neurosci. Lett. 381 (3), 309–313.
Doppelmayr, M., Klimesch, W., Hödlmoser, K., Sauseng, P., Gruber, W., 2005b. Intelligence related upper alpha desynchronization in a semantic memory task. Brain Res. Bull. 66 (2), 171–177.
Dunst, B., Benedek, M., Jauk, E., Bergner, S., Koschutnig, K., Sommer, M., et al., 2014. Neural efficiency as a function of task demands. Intelligence 42, 22–30.
Duval, R., 2006. A cognitive analysis of problems of comprehension in a learning of mathematics. Educ. Stud. Math. 61 (1), 103–131.
Eisenberg, T., 2002. Functions and associated learning difficulties. In: Tall, D. (Ed.), Advanced Mathematical Thinking. Springer, pp. 140–152.
Elia, I., Gagatsis, A., Demetriou, A., 2007. The effects of different modes of representation on the solution of one-step additive problems. Learn. InStruct. 17 (6), 658–672.
Gagatsis, A., Shiakalli, M., 2004. Ability to translate from one representation of the concept of function to another and mathematical problem solving. Educ. Psychol. 24 (5), 645–657.
Grabner, R.H., De Smedt, B., 2011. Neurophysiological evidence for the validity of verbal strategy reports in mental arithmetic. Biol. Psychol. 87 (1), 128–136.
Grabner, R.H., Brunner, C., Lorenz, V., Vogel, S.E., De Smedt, B., 2022. Fact retrieval or compacted counting in arithmetic – a neurophysiological investigation of two hypotheses. J. Exp. Psychol. Learn. Mem. Cognit. 48 (2), 199–212.
Grabner, R.H., Fink, A., Stipacek, A., Neuper, C., Neubauer, A.C., 2004. Intelligence and working memory systems: evidence of neural efficiency in alpha band ERD. Cognit. Brain Res. 20 (2), 212–225.
Grabner, R.H., Neubauer, A.C., Stern, E., 2006. Superior performance and neural efficiency: the impact of intelligence and expertise. Brain Res. Bull. 69 (4), 422–439.
Gratton, G., Coles, M.G.H., Donchin, E., 1983. A new method for off-line removal of ocular artifact. Electroencephalogr. Clin. Neurophysiol. 55 (4), 468–484.
Hiebert, J., Carpenter, T.P., 1992. Learning and teaching with understanding. In: Grouws, D.A. (Ed.), Handbook of Research on Mathematics Teaching and Learning. Macmillan, New York, pp. 65–97.
Janvier, C., 1987. Translation processes in mathematics education. In: Janvier, C. (Ed.), Problems of Representation in the Teaching and Learning of Mathematics. Lawrence Erlbaum, pp. 27–32.
Jung, R.E., Haier, R.J., 2007. The Parieto-Frontal Integration Theory (P-FIT) of intelligence: converging neuroimaging evidence. Behav. Brain Sci. 30 (2), 135.
Kaput, J.J., 1998. Representations, inscriptions, descriptions and learning: a kaleidoscope of windows. J. Math. Behav. 17 (2), 265–281.
Keig, P.F., Rubba, P.A., 1993. Translation of representations of the structure of matter and its relationship to reasoning, gender, spatial reasoning, and specific prior knowledge. J. Res. Sci. Teach. 30 (8), 883–903.
Kelly, A.C., Garavan, H., 2005. Human functional neuroimaging of brain changes associated with practice. Cerebr. Cortex 15 (8), 1089–1102.
Klimesch, W., 2012. Alpha-band oscillations, attention, and controlled access to stored information. Trends Cognit. Sci. 16 (12), 606–617.
Klimesch, W., Schack, B., Sauseng, P., 2005. The functional significance of theta and upper alpha oscillations. Exp. Psychol. 52 (2), 99–108.
Koshy, V., Ernest, P., Casey, R., 2009. Mathematically gifted and talented learners: theory and practice. Int. J. Math. Educ. Sci. Technol. 40 (2), 213–228.
Ku, Y., Hong, B., Gao, X., Gao, S., 2010. Spectra-temporal patterns underlying mental addition: an ERP and ERD/ERS study. Neurosci. Lett. 472 (1), 5–10.
Lee, K., Ng, S.F., Bull, R., 2017. Learning and solving more complex problems: the roles of working memory, updating, and prior skills for general mathematical achievement and algebra. In: Acquisition of Complex Arithmetic Skills and Higher-Order Mathematics Concepts. Academic Press, , pp. 197–220.
Leikin, M., Paz-Baruch, N., Leikin, R., 2013a. Memory abilities in generally gifted and excelling-in-mathematics adolescents. Intelligence 41 (5), 566–578.
Leikin, M., Waisman, I., Leikin, R., 2013b. How brain research can contribute to the evaluation of mathematical giftedness. Psychological Test and Assessment Modeling 55 (4), 415.
Leikin, R., 2020. Giftedness and high ability in mathematics. Encyclopedia of mathematics education 315–325.
Leikin, R., Leikin, M., 2009. *Multidimensional Examination Of Mathematical Genius As Compared To Mathematically Gifted And Non-gifted Children*. Research Proposal Granted by. The John Templeton Foundation (Grant # 1447).
Leikin, R., Lev, M., 2013. Mathematical creativity in generally gifted and mathematically excelling adolescents: what makes the difference? ZDM – Mathematics Education 45 (2), 183–197.
Leikin, R., Waisman, I., Leikin, M., 2016. Does solving insight-based problems differ from solving learning-based problems? Some evidence from an ERP study. ZDM – Mathematics Education 48 (3), 305–319.
Leinhardt, G., Zaslavsky, O., Stein, M.K., 1990. Functions, graphs, and graphing: tasks, learning, and teaching. Rev. Educ. Res. 60 (1), 1–64.
Lubinski, D., Benbow, C.P., 2006. Study of mathematically precocious youth after 35 years: uncovering antecedents for the development of math-science expertise. Perspect. Psychol. Sci. I (4), 316–345.
Marko, M., Cimrová, B., Riečanský, I., 2019. Neural theta oscillations support semantic memory retrieval. Sci. Rep. 9 (1), 1–10.
Morrison, W., Collymore, R., Saul, M., Paul, K., 2006. Remember to Read the Question: A Thinking Students' Guide to the SAT and beyond. Kendall & Hunt.
Neubauer, A.C., Fink, A., 2009. Intelligence and neural efficiency. Neurosci. Biobehav. Rev. 33 (7), 1004–1023.
Neubauer, A.C., Sange, G., Pfurtscheller, G., 1999. Psychometric intelligence and event-related desynchronization during performance of a letter matching task. In: Pfurtscheller, G., Lopes da Silva, F.H. (Eds.), Event-Related Desynchronization. Handbook of Electroencephalography. And Clinical Neurophysiology, vol. 6. Elsevier BV, pp. 219–231.
Nussbaumer, D., Grabner, R.H., Stern, E., 2015. Neural efficiency in working memory tasks: the impact of task demand. Intelligence 50, 196–208.
Paz-Baruch, N., Leikin, M., Aharon-Peretz, J., Leikin, R., 2014. Speed of information processing in generally gifted and excelling-in-mathematics adolescents. High. Abil. Stud. 25 (2), 143–167.
Paz-Baruch, N., Leikin, R., Leikin, M., 2016. Visual processing in generally gifted and mathematically excelling adolescents. J. Educ. Gift. 39 (3), 237–258.
Paz-Baruch, N., Leikin, M., Leikin, R., 2022. Not any gifted is an expert in mathematics and not any expert in mathematics is gifted. Gifted and Talented International 37 (1), 25–41.
Pfurtscheller, G., Da Silva, F.L., 1999. Event-related EEG/MEG synchronization and desynchronization: basic principles. Clin. Neurophysiol. 110 (11), 1842–1857.
Polya, G., 1973. How to Solve it. A New Aspect of Mathematical Method. Princeton University Press, Princeton.

Ponte, J.P.D., 1992. The history of the concept of function and some educational implications. Math. Educat. 3 (2), 3–8.
Popescu, T., Sader, E., Schaer, M., Thomas, A., Terhune, D.B., Dowker, A., et al., 2019. The brain-structural correlates of mathematical expertise. Cortex 114, 140–150.
Prescott, J., Gavrilescu, M., Cunnington, R., O'Boyle, M.W., Egan, G.F., 2010. Enhanced brain connectivity in math-gifted adolescents: an fMRI study using mental rotation. Cognit. Neurosci. 1 (4), 277–288.
Raven, J., Raven, J.C., Court, J.H., 2000. Manual for Raven's Progressive Matrices and Vocabulary Scales. Oxford Psychologists.
Salisbury, D.F., Taylor, G., 2012. Semantic priming increases left hemisphere theta power and intertrial phase synchrony. Psychophysiology 49 (3), 305–311.
Schneider, W., Eschman, A., Zuccolotto, A., 2002. *E-Prime Computer Software* (Version 1.0). Psychology Software Tools.
Schönborn, K.J., Bögeholz, S., 2009. Knowledge transfer in biology and translation across external representations: experts' views and challenges for learning. Int. J. Sci. Math. Educ. 7 (5), 931–955.
Silverman, L.K., 2009. The measurement of giftedness. In: Shavinina, L.V. (Ed.), International handbook on giftedness. Springer, pp. 947–970.
Taub, G.E., Keith, T.Z., Floyd, R.G., McGrew, K.S., 2008. Effects of general and broad cognitive abilities on mathematics achievement. Sch. Psychol. Q. 23 (2), 187–198.
Threlfall, J., Hargreaves, M., 2008. The problem-solving methods of mathematically gifted and older average-attaining students. High. Abil. Stud. 19 (1), 83–98.
Van Heuvelen, A., Zou, X., 2001. Multiple representations of work–energy processes. Am. J. Phys. 69 (2), 184–194.
Waisman, I., Leikin, M., Shaul, S., Leikin, R., 2014. Brain activity associated with translation between graphical and symbolic representations of functions in generally gifted and excelling in mathematics adolescents. Int. J. Sci. Math. Educ. 12 (3), 669–696.
Young, J.W., Kobrin, J.L., 2001. Differential validity, differential prediction, and college admission testing: a comprehensive review and analysis. Research report No. 2001–6.
Zohar, A., 1990. Mathematical Reasoning Ability: its Structure, and Some Aspects of its Genetic Transmission. Unpublished Doctoral Dissertation, Hebrew University, Jerusalem.